```
1   FUNCTION_BLOCK PR_5by3CylModWoRdlck
2   VAR_INPUT
3       iResetModule : BOOL ;  // Reset if 1
4       iResetFaults : BOOL ;  // Reset Fault CMD
5       iEnergizeAtPwrOn : BOOL ;  // Energize Poweron Input
6       iWorkCmd : BOOL ;  // Cyl Work Postion CMD
7       iHomeCmd : BOOL ;  // Cyl Home Postion CMD
8       iWorkSen : BOOL ;  // Cyl Work Postion Input
9       iHomeSen : BOOL ;  // Cyl Home Postion Input
10      iWorkCleartoMove : BOOL ;  // connect work position interlock condition
    healthy
11      iHomeCleartoMove : BOOL ;  // connect home position interlock condition
    healthy
12      iWorkPositionLeftOption : BOOL ;  // It is the selection of the to keep
    work solenoid coil on after reaching work position
13      iHomePositionLeftOption : BOOL ;  // It is the selection of the to keep
    home solenoid coil on after reaching home position
14      iTimeoutValue : TIME ;  // Time Required E.g T#500MS,T#2S
15      iWorkSensDebounceVal : TIME ;  // Time Required E.g T#500MS,T#2S
16      iHomeSensDebounceVal : TIME ;  // Time Required E.g T#500MS,T#2S
17  END_VAR
18  VAR_OUTPUT
19      qWorkSol : BOOL ;  // Cyl Work Position Sol Output
20      qHomeSol : BOOL ;  // Cyl Home Position Sol Output
21      qWorkOver : BOOL ;  // 1 if Cyl at Work Position
22      qHomeOver : BOOL ;  // 1 if Cyl at Home Position
23      qModError : BOOL ;  // Module Error
24      qErrBothSensorOn : BOOL ;  // 1 If both WorkSen and HomeSen inputs are high
    at the same time.
25      qErrHomeInterlock : BOOL ;  // 1 If homeCleartoMove is low and home command
    is active.
26      qErrHomeSenNotLeft : BOOL ;  // 1 If home position sensor did not turn off
    within defined time
27      qErrHomeTout : BOOL ;  // 1 If cylinder did not reach home position within
    defined time
28      qErrWorkInterlock : BOOL ;  // 1 If WorkCleartoMove is low and work command
    is active.
29      qErrWorkSenNotLeft : BOOL ;  // 1 If work position sensor did not turn off
    within defined time
30      qErrWorkTout : BOOL ;  // 1 If cylinder did not reach work position within
    defined time
31      qMsgWorkPosLost : BOOL ;  // 1 If work position achieved and lost its
    position without home
32      qMsgHomePosLost : BOOL ;  // 1 If home position achieved and lost its
    position without work
33      qMsgBothCmdOn : BOOL ;  // 1 if Home & work position command on same time
34  END_VAR
35  VAR
36      sFltRstPls : BOOL ;  // Stat bit for positive edge of reset fault input
37      sFltEdge : R_TRIG ;  // Positive edge of fault reset input
38      sWorkPosMemBit : BOOL ;  // Stat bit for work position memory bit
39      sHomePosMemBit : BOOL ;  // Stat bit for Home position memory bit
```

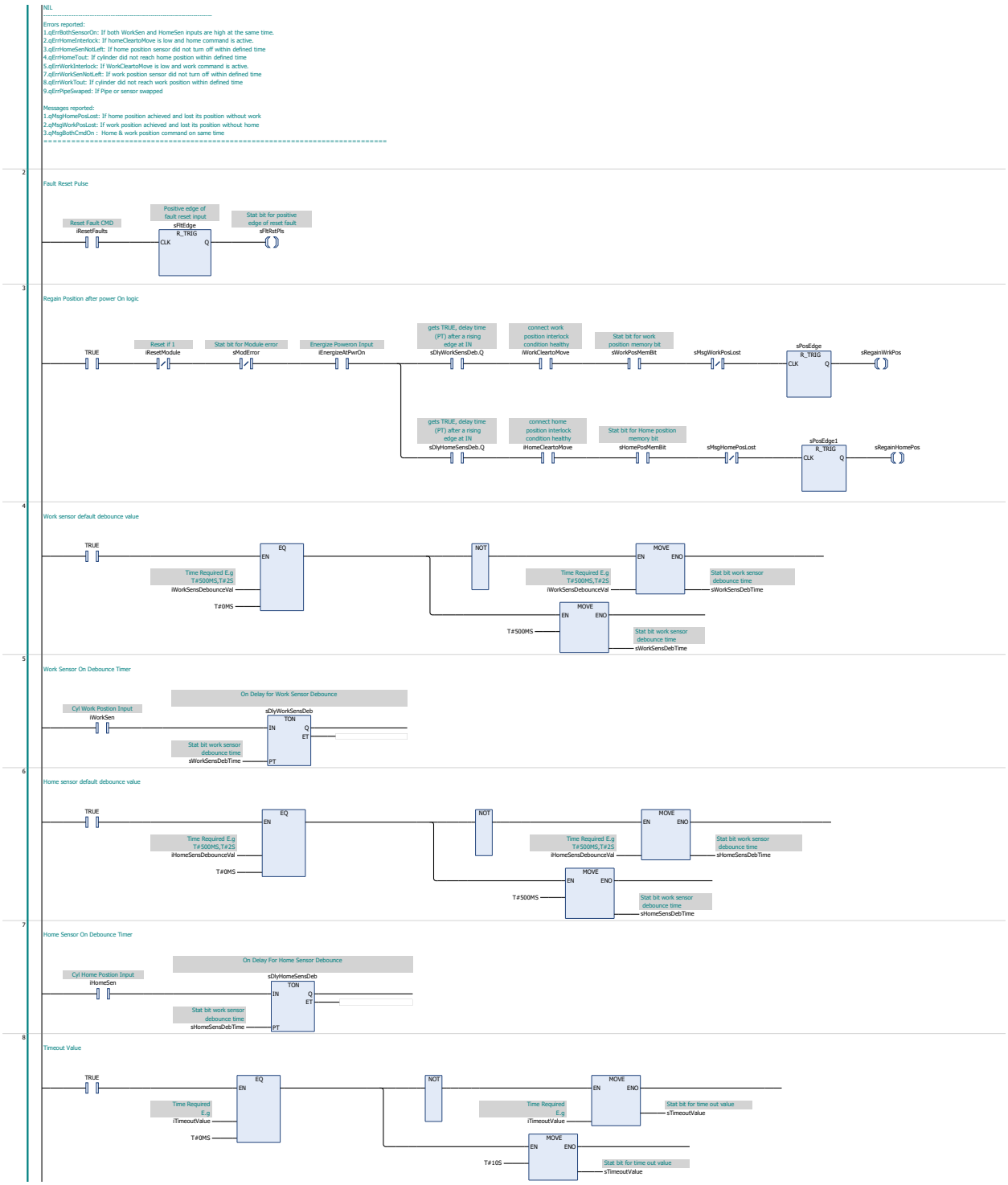
```
40          sModError  :  BOOL ;   // Stat bit for Module error
41          sWorkSensDebTime  :  TIME ;   // Stat bit work sensor debounce time
42          sHomeSensDebTime  :  TIME ;   // Stat bit work sensor debounce time
43          sTimeoutValue  :  TIME ;   // Stat bit for time out value
44          sElapsedTime  :  TIME ;   // Stat bit for time elapsed
45          sDlyWorkSensDeb  :  TON ;   // On Delay for Work Sensor Debounce
46          sDlyHomeSensDeb  :  TON ;   // On Delay For Home Sensor Debounce
47          sDlyTimeOut  :  TON ;   // On Delay Time Out timer
48          sWorkSol  :  BOOL ;
49          sErrWorkTout  :  BOOL ;
50          sHomeSol  :  BOOL ;
51          sErrHomeTout  :  BOOL ;
52          sErrBothSensorOn  :  BOOL ;
53          sErrWorkInterlock  :  BOOL ;
54          sErrHomeInterlock  :  BOOL ;
55          sErrWorkSenNotLeft  :  BOOL ;
56          sErrHomeSenNotLeft  :  BOOL ;
57          sMsgHomePosLost  :  BOOL ;
58          sMsgWorkPosLost  :  BOOL ;
59          sPosEdge  :  R_TRIG ;
60          sPosEdge1  :  R_TRIG ;
61          sRegainWrkPos  :  BOOL ;
62          sRegainHomePos  :  BOOL ;
63          sBothCmdOn  :  BOOL ;
64          sWorkPositionLeft  :  BOOL ;
65          sHomePositionLeft  :  BOOL ;
66      END_VAR
67
```

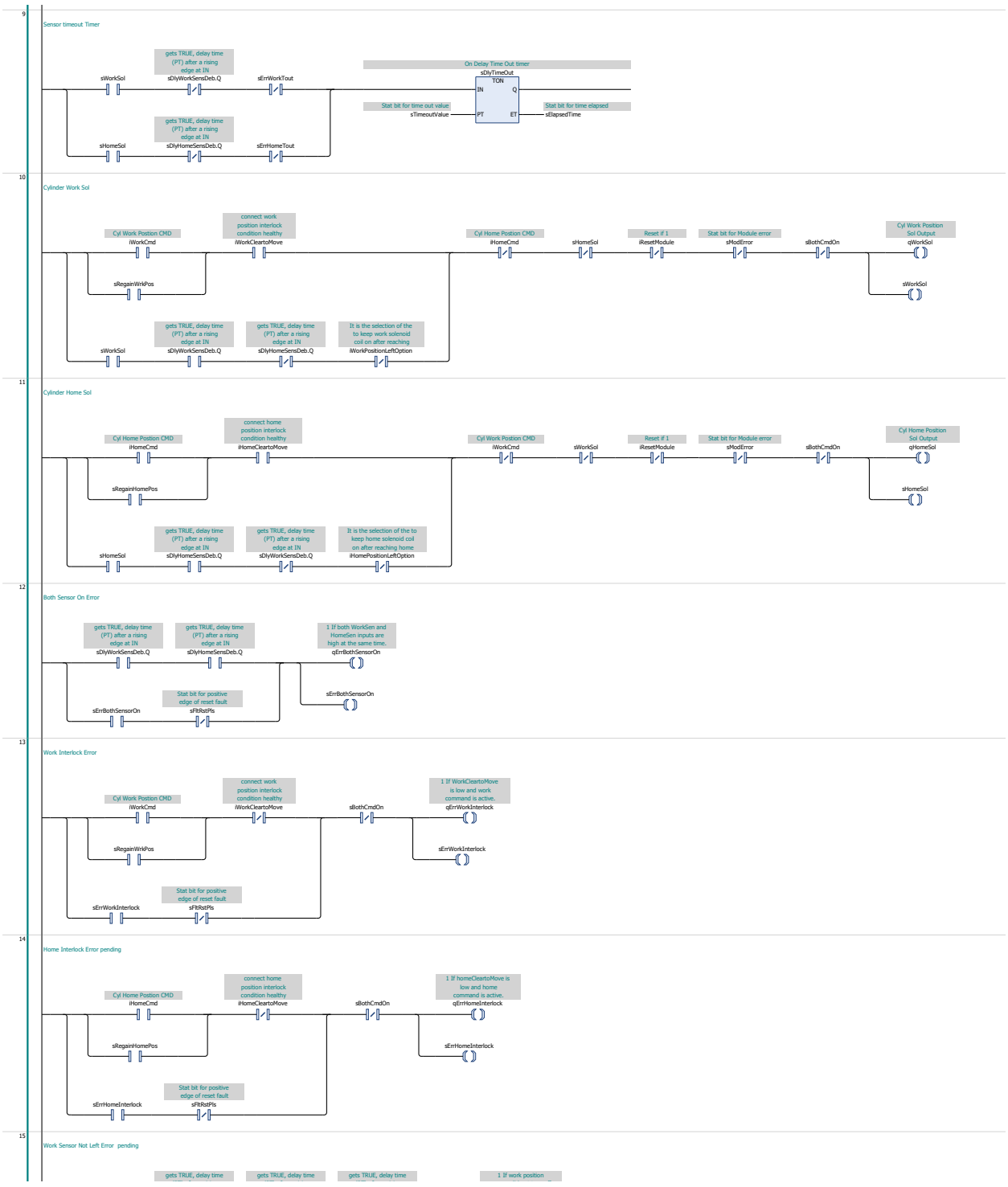
```
1
(*
================================================================================
Company: Precision Automation & Robotics India Ltd.
--------------------------------------------------------------------------------

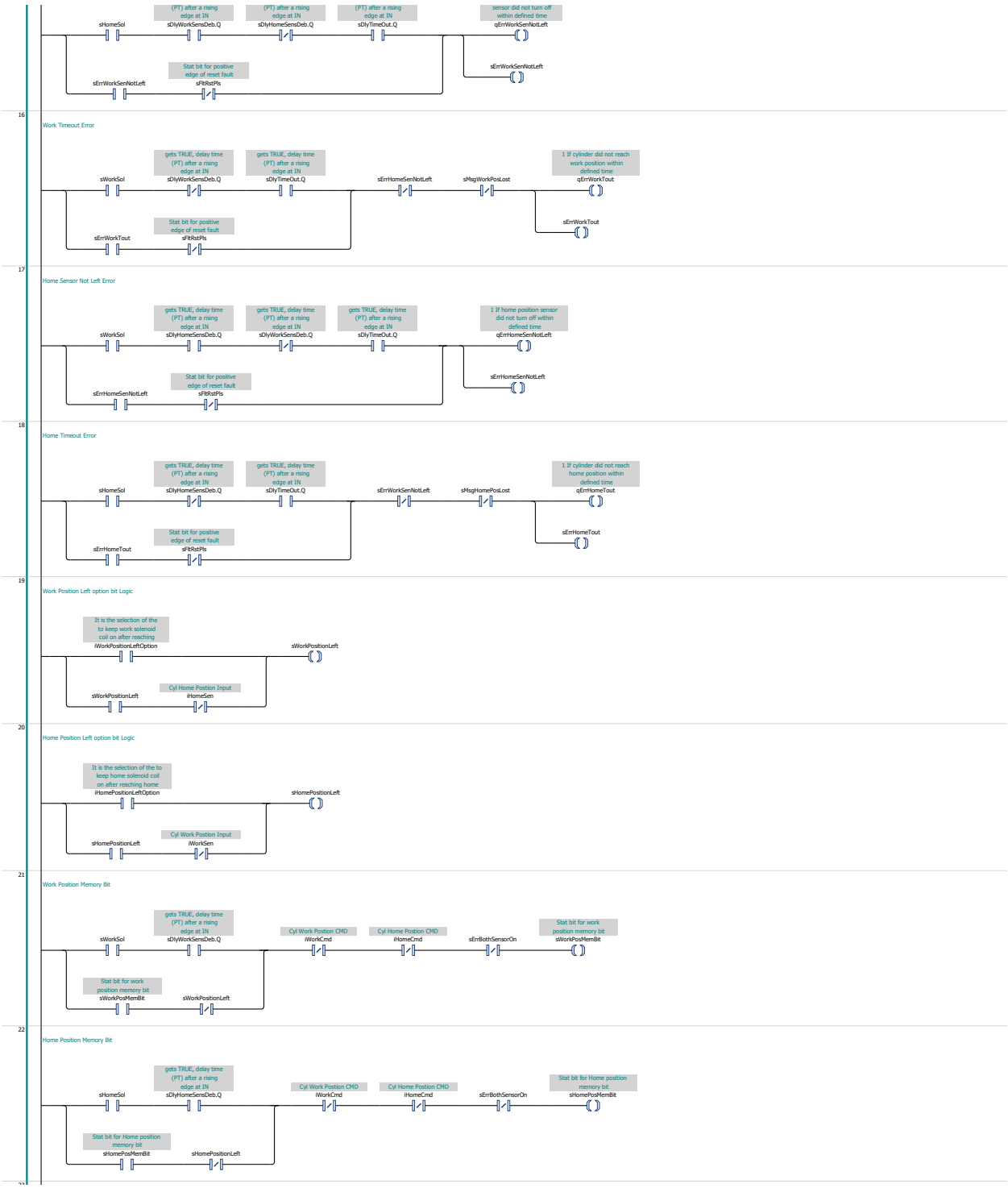
Item Code:SM0101022005
Author/Created by: Amiya Das
Reviewed/Checked by: Atul Patwardhan
First Release on: 20-01-2017
Latest update on:27-01-2025
--------------------------------------------------------------------------------
Tested with:Simulator
Engineering/Editor: TIA Portal 15 & Higher
--------------------------------------------------------------------------------
Restrictions: No
Dependencies/Requirements: NA
--------------------------------------------------------------------------------
Change Log:
Ver.    Date                Changed by              Details of changes
0.2     20-01-2017          Amiya Das               Second release version
0.3     27-01-2025          Sujeet Rote             qMsgBothCmdOn message is added
0.4     29-01-2025          Sujeet Rote             Work & home position left option bit
--------------------------------------------------------------------------------
Functionality: This block is used to control a simple motion with one work
position output, one home position  output, one work position sensor and one
home position sensor.
Inputs
iResetModule  Bool
iResetFaults  Bool
iEnergizeAllPwrOnBool
iWorkCmd Bool
iHomeCmd  Bool
iWorkSen  Bool
iHomeSen Bool
iWorkCleartoMove  Bool
iHomeCleartoMove  Bool
iWorkPositionLeftOption  Bool
iHomePositionLeftOption Bool
iTimeoutValue Time
iWorkSensDebounceVal  Time
iHomeSensDebounceVal  Time

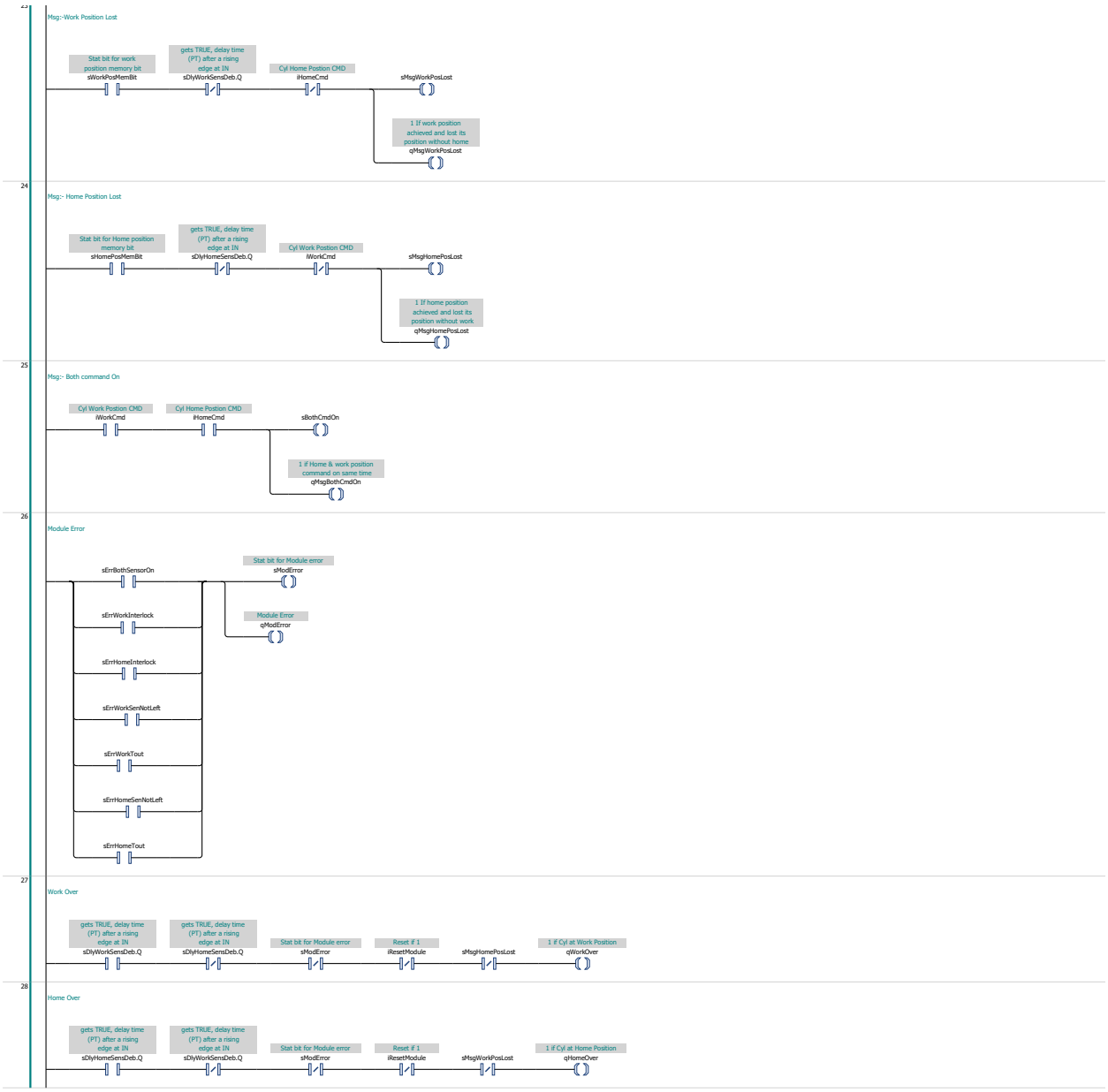
Outputs
qWorkSol  Bool
qHomeSol Bool
qWorkOver  Bool
qHomeOver  Bool
qErrBothSensorOn  Bool
qErrHomeInterlock  Bool
qErrHomeSenNotLeft Bool
qErrHomeTout  Bool
qErrWorkInterlock  Bool
qErrWorkSenNotLeft  Bool
qErrWorkTout Bool
qMsgWorkPosLost  Bool
qMsgHomePosLost  Bool
qMsgBothCmdOn Bool

In-outs
--------------------------------------------------------------------------------
*)
```


NIL
Errors reported:
1.qErrBothSensorOn: If both WorkSen and HomeSen inputs are high at the same time.
2.qErrHomeInterlock: If homeCleartoMove is low and home command is active.
3.qErrHomeSenNotLeft: If home position sensor did not turn off within defined time
4.qErrHomeTout: If cylinder did not reach home position within defined time
5.qErrWorkInterlock: If WorkCleartoMove is low and work command is active.
7.qErrWorkSenNotLeft: If work position sensor did not turn off within defined time
8.qErrWorkTout: If cylinder did not reach work position within defined time
9.qErrPipeSwaped: If Pipe or sensor swapped
Messages reported:
1.qMsgHomePosLost: If home position achieved and lost its position without work
2.qMsgWorkPosLost: If work position achieved and lost its position without home
3.qMsgBothCmdOn : Home & work position command on same time
2
Fault Reset Pulse
Reset Fault CMD
iResetFaults
Positive edge of fault reset input
sFltEdge
R_TRIG
CLK
Q
Stat bit for positive edge of reset fault
sFltRstPs
3
Regain Position after power On logic
TRUE
Reset if 1
iResetModule
Stat bit for Module error
sModError
Energize Poweron Input
iEnergizeAtPwrOn
gets TRUE, delay time (PT) after a rising edge at IN
sDlyWorkSensDeb.Q
connect work position interlock condition healthy
iWorkCleartoMove
Stat bit for work position memory bit
sWorkPosMemBit
sMsgWorkPosLost
sPosEdge
R_TRIG
CLK
Q
sRegainWrkPos
gets TRUE, delay time (PT) after a rising edge at IN
sDlyHomeSensDeb.Q
connect home position interlock condition healthy
iHomeCleartoMove
Stat bit for Home position memory bit
sHomePosMemBit
sMsgHomePosLost
sPosEdge1
R_TRIG
CLK
Q
sRegainHomePos
4
Work sensor default debounce value
TRUE
EQ
EN
Time Required E.g T#500MS,T#2S
iWorkSensDebounceVal
T#0MS
NOT
MOVE
EN
ENO
Time Required E.g T#500MS,T#2S
iWorkSensDebounceVal
Stat bit work sensor debounce time
sWorkSensDebTime
T#500MS
Stat bit work sensor debounce time
sWorkSensDebTime
5
Work Sensor On Debounce Timer
Cyl Work Postion Input
iWorkSen
On Delay for Work Sensor Debounce
sDlyWorkSensDeb
TON
IN
Q
ET
PT
Stat bit work sensor debounce time
sWorkSensDebTime
6
Home sensor default debounce value
TRUE
EQ
EN
Time Required E.g T#500MS,T#2S
iHomeSensDebounceVal
T#0MS
NOT
MOVE
EN
ENO
Time Required E.g T#500MS,T#2S
iHomeSensDebounceVal
Stat bit work sensor debounce time
sHomeSensDebTime
T#500MS
Stat bit work sensor debounce time
sHomeSensDebTime
7
Home Sensor On Debounce Timer
Cyl Home Postion Input
iHomeSen
On Delay For Home Sensor Debounce
sDlyHomeSensDeb
TON
IN
Q
ET
PT
Stat bit work sensor debounce time
sHomeSensDebTime
8
Timeout Value
TRUE
EQ
EN
Time Required E.g
iTimeoutValue
T#0MS
NOT
MOVE
EN
ENO
Time Required E.g
iTimeoutValue
Stat bit for time out value
sTimeoutValue
T#10S
Stat bit for time out value
sTimeoutValue


9
Sensor timeout Timer
gets TRUE, delay time (PT) after a rising edge at IN
On Delay Time Out timer
sDlyTimeOut
TON
sWorkSol
sDlyWorkSensDeb.Q
sErrWorkTout
IN
Q
Stat bit for time out value
Stat bit for time elapsed
sTimeoutValue
PT
ET
sElapsedTime
gets TRUE, delay time (PT) after a rising edge at IN
sHomeSol
sDlyHomeSensDeb.Q
sErrHomeTout
10
Cylinder Work Sol
Cyl Work Postion CMD
connect work position interlock condition healthy
Cyl Home Postion CMD
Reset if 1
Stat bit for Module error
Cyl Work Position Sol Output
iWorkCmd
iWorkClearToMove
iHomeCmd
sHomeSol
iResetModule
sModError
sBothCmdOn
qWorkSol
sRegainWrkPos
sWorkSol
gets TRUE, delay time (PT) after a rising edge at IN
gets TRUE, delay time (PT) after a rising edge at IN
It is the selection of the to keep work solenoid coil on after reaching
sDlyHomeSensDeb.Q
iWorkPositionLeftOption
11
Cylinder Home Sol
Cyl Home Postion CMD
connect home position interlock condition healthy
Cyl Work Postion CMD
Reset if 1
Stat bit for Module error
Cyl Home Position Sol Output
iHomeCmd
iHomeClearToMove
iWorkCmd
sWorkSol
iResetModule
sModError
sBothCmdOn
qHomeSol
sRegainHomePos
sHomeSol
gets TRUE, delay time (PT) after a rising edge at IN
gets TRUE, delay time (PT) after a rising edge at IN
It is the selection of the to keep home solenoid coil on after reaching home
sDlyWorkSensDeb.Q
iHomePositionLeftOption
12
Both Sensor On Error
gets TRUE, delay time (PT) after a rising edge at IN
gets TRUE, delay time (PT) after a rising edge at IN
1 If both WorkSen and HomeSen inputs are high at the same time.
sDlyWorkSensDeb.Q
sDlyHomeSensDeb.Q
qErrBothSensorOn
Stat bit for positive edge of reset fault
sErrBothSensorOn
sFltRstPls
13
Work Interlock Error
Cyl Work Postion CMD
connect work position interlock condition healthy
1 If WorkClearToMove is low and work command is active.
iWorkCmd
iWorkClearToMove
sBothCmdOn
qErrWorkInterlock
sRegainWrkPos
sErrWorkInterlock
Stat bit for positive edge of reset fault
sErrWorkInterlock
sFltRstPls
14
Home Interlock Error pending
Cyl Home Postion CMD
connect home position interlock condition healthy
1 If homeClearToMove is low and home command is active.
iHomeCmd
iHomeClearToMove
sBothCmdOn
qErrHomeInterlock
sRegainHomePos
sErrHomeInterlock
Stat bit for positive edge of reset fault
sErrHomeInterlock
sFltRstPls
15
Work Sensor Not Left Error pending
gets TRUE, delay time
gets TRUE, delay time
gets TRUE, delay time
1 If work position


(PT) after a rising edge at IN
sHomeSol
sDlyWorkSensDeb.Q
(PT) after a rising edge at IN
sDlyHomeSensDeb.Q
(PT) after a rising edge at IN
sDlyTimeOut.Q
sensor did not turn off within defined time
qErrWorkSenNotLeft
sErrWorkSenNotLeft
Stat bit for positive edge of reset fault
sFltRstPls
sErrWorkSenNotLeft
16
Work Timeout Error
gets TRUE, delay time (PT) after a rising edge at IN
sWorkSol
sDlyWorkSensDeb.Q
gets TRUE, delay time (PT) after a rising edge at IN
sDlyTimeOut.Q
sErrHomeSenNotLeft
sMsgWorkPosLost
1 If cylinder did not reach work position within defined time
qErrWorkTout
sErrWorkTout
Stat bit for positive edge of reset fault
sErrWorkTout
sFltRstPls
17
Home Sensor Not Left Error
gets TRUE, delay time (PT) after a rising edge at IN
sWorkSol
sDlyHomeSensDeb.Q
gets TRUE, delay time (PT) after a rising edge at IN
sDlyWorkSensDeb.Q
gets TRUE, delay time (PT) after a rising edge at IN
sDlyTimeOut.Q
1 If home position sensor did not turn off within defined time
qErrHomeSenNotLeft
sErrHomeSenNotLeft
Stat bit for positive edge of reset fault
sErrHomeSenNotLeft
sFltRstPls
18
Home Timeout Error
gets TRUE, delay time (PT) after a rising edge at IN
sHomeSol
sDlyHomeSensDeb.Q
gets TRUE, delay time (PT) after a rising edge at IN
sDlyTimeOut.Q
sErrWorkSenNotLeft
sMsgHomePosLost
1 If cylinder did not reach home position within defined time
qErrHomeTout
sErrHomeTout
Stat bit for positive edge of reset fault
sErrHomeTout
sFltRstPls
19
Work Position Left option bit Logic
It is the selection of the to keep work solenoid coil on after reaching
iWorkPositionLeftOption
sWorkPositionLeft
sWorkPositionLeft
Cyl Home Postion Input
iHomeSen
20
Home Position Left option bit Logic
It is the selection of the to keep home solenoid coil on after reaching home
iHomePositionLeftOption
sHomePositionLeft
sHomePositionLeft
Cyl Work Postion Input
iWorkSen
21
Work Position Memory Bit
gets TRUE, delay time (PT) after a rising edge at IN
sWorkSol
sDlyWorkSensDeb.Q
Cyl Work Postion CMD
iWorkCmd
Cyl Home Postion CMD
iHomeCmd
sErrBothSensorOn
Stat bit for work position memory bit
sWorkPosMemBit
Stat bit for work position memory bit
sWorkPosMemBit
sWorkPositionLeft
22
Home Position Memory Bit
gets TRUE, delay time (PT) after a rising edge at IN
sHomeSol
sDlyHomeSensDeb.Q
Cyl Work Postion CMD
iWorkCmd
Cyl Home Postion CMD
iHomeCmd
sErrBothSensorOn
Stat bit for Home position memory bit
sHomePosMemBit
Stat bit for Home position memory bit
sHomePosMemBit
sHomePositionLeft


Msg:-Work Position Lost
Stat bit for work position memory bit
sWorkPosMemBit
gets TRUE, delay time (PT) after a rising edge at IN
sDlyWorkSensDeb.Q
Cyl Home Postion CMD
iHomeCmd
sMsgWorkPosLost
1 If work position achieved and lost its position without home
qMsgWorkPosLost
Msg:- Home Position Lost
Stat bit for Home position memory bit
sHomePosMemBit
gets TRUE, delay time (PT) after a rising edge at IN
sDlyHomeSensDeb.Q
Cyl Work Postion CMD
iWorkCmd
sMsgHomePosLost
1 If home position achieved and lost its position without work
qMsgHomePosLost
Msg:- Both command On
Cyl Work Postion CMD
iWorkCmd
Cyl Home Postion CMD
iHomeCmd
sBothCmdOn
1 if Home & work position command on same time
qMsgBothCmdOn
Module Error
sErrBothSensorOn
Stat bit for Module error
sModError
sErrWorkInterlock
Module Error
qModError
sErrHomeInterlock
sErrWorkSenNotLeft
sErrWorkTout
sErrHomeSenNotLeft
sErrHomeTout
Work Over
gets TRUE, delay time (PT) after a rising edge at IN
sDlyWorkSensDeb.Q
gets TRUE, delay time (PT) after a rising edge at IN
sDlyHomeSensDeb.Q
Stat bit for Module error
sModError
Reset if 1
iResetModule
sMsgHomePosLost
1 if Cyl at Work Position
qWorkOver
Home Over
gets TRUE, delay time (PT) after a rising edge at IN
sDlyHomeSensDeb.Q
gets TRUE, delay time (PT) after a rising edge at IN
sDlyWorkSensDeb.Q
Stat bit for Module error
sModError
Reset if 1
iResetModule
sMsgWorkPosLost
1 if Cyl at Home Position
qHomeOver